

In Class ExerciseJanuary 28, 2026

Suppose we observe housing data $\{(x_i, y_i)\}_{i=1}^n$, where:

- x_i denotes the size of house i (in square feet),
- y_i denotes the house price (in thousands of dollars).

We consider the linear regression model

$$y_i = \beta_0 + \beta_1 x_i + \varepsilon_i,$$

where β_0 is the intercept and β_1 is the slope.

Part (a) Write the optimization problem that estimates (β_0, β_1) by minimizing the *sum of absolute errors*:

Part (b) Is this optimization problem linear? Briefly explain why or why not.